

An Explainable Multi-Mode Board Game Recommendation System Using Content Similarity, Sentiment Analysis, and Popularity Signals

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Abstract—Board games have experienced significant growth in popularity, resulting in increasingly large and complex game catalogs that make game discovery difficult for players. Traditional recommendation approaches often rely heavily on collaborative filtering, which suffers from sparsity and cold-start limitations when user interaction data is limited. To address these challenges, this study proposes an explainable hybrid board game recommendation system that combines content similarity, sentiment analysis, and popularity signals to support multiple recommendation modes.

The proposed system integrates textual game descriptions, gameplay traits, user review sentiment, and popularity-based information to generate recommendations through title-based, trait-based, and combined recommendation modes. Content representations were generated using TF-IDF and sentence-transformer embeddings, while sentiment scores were derived from transformer-based review analysis. The system was evaluated using a filtered subset of the BoardGameGeek dataset containing over 21,000 board games and millions of user ratings and reviews.

The system was evaluated on 300 BoardGameGeek users with Recall@K and NDCG@K under a holdout protocol, alongside a popularity baseline, a content-only baseline, and a sentiment ablation. Under this protocol, whose held-out games are dominated by the most-rated titles, the popularity baseline was clearly strongest (Recall@10 of 0.230), while the three proposed modes reached Recall@10 between 0.020 and 0.037, the combined mode being best among them. Counting only held-out games outside the 100 most-rated titles, the popularity baseline retrieved none, whereas the proposed modes retrieved a small share (Recall@10 of 0.015–0.022).

The findings indicate that offline accuracy on this dataset is driven largely by popularity, and that the value of the proposed framework lies in explainable, similarity-driven discovery beyond the most popular games rather than in raw retrieval accuracy. The paper reports these results candidly and identifies the evaluation and modelling changes needed to strengthen them.

Index Terms—Recommender Systems, Board Game Recommendation, Explainable Recommendation, Content-Based Filtering, Sentiment Analysis, Hybrid Recommendation, BoardGameGeek

I. INTRODUCTION

Board games have experienced significant growth in popularity over the past decade, resulting in increasingly large and complex game catalogs on online platforms such as BoardGameGeek (BGG). As the number of available games continues to expand, users often face difficulties in discovering

games that match their preferences, gameplay styles, and thematic interests. This creates a recommendation challenge in which systems must balance personalization, relevance, diversity, and explainability.

Traditional recommendation systems commonly rely on collaborative filtering techniques, which generate recommendations based on user interaction patterns and rating similarities. Although collaborative filtering has shown strong performance in many domains, it suffers from several important limitations. One major issue is the cold-start problem, where newly added games or users with limited interaction history cannot be effectively recommended. In addition, sparse rating distributions may reduce recommendation quality and system reliability.

Content-based recommendation approaches attempt to address some of these limitations by utilizing item attributes such as descriptions, mechanics, categories, and themes. However, content-only systems often fail to capture qualitative user opinions and emotional responses toward games. Meanwhile, popularity-based approaches may over-recommend highly popular games while ignoring niche preferences and personalized interests.

To address these challenges, this paper proposes an explainable hybrid board game recommendation system that combines multiple recommendation signals into a unified framework. The proposed system integrates content similarity, sentiment analysis derived from user reviews, and popularity information to generate interpretable recommendations, and evaluates candidly how much each signal contributes. Furthermore, the system supports multiple recommendation modes, including title-based recommendations, trait-based recommendations, and combined hybrid recommendations.

The primary contributions of this paper are as follows:

- Development of a hybrid recommendation framework combining content similarity, sentiment signals, and popularity metrics.
- Implementation of explainable recommendation outputs that provide interpretable recommendation reasoning.
- Support for multiple recommendation modes, including title-based, trait-based, and combined recommendations.
- Evaluation of recommendation performance using Recall@K and NDCG@K metrics under different recommendation scenarios.

The remainder of this paper is organized as follows. Section II reviews related work in recommendation systems and sentiment-aware recommendation approaches. Section III describes the proposed methodology and recommendation architecture. Section IV presents the experimental evaluation and performance analysis. Section V discusses the findings, limitations, and implications of the proposed system. Finally, Section VI concludes the paper and suggests directions for future work.

II. RELATED WORK

Recommendation systems have been widely studied as a solution for helping users discover relevant items in large information spaces. Collaborative filtering is one of the most established recommendation approaches and relies on user-item interaction patterns to identify similarities between users or items. Su and Khoshgoftaar [1] provide a broad survey of collaborative filtering techniques, highlighting their effectiveness as well as their limitations, including rating sparsity and cold-start problems. Herlocker et al. [2] further emphasize the importance of proper evaluation methods when assessing collaborative filtering systems.

Content-based recommendation systems address some of these limitations by recommending items based on item attributes rather than user interaction histories. Pazzani and Billsus [3] describe how content-based systems use textual and metadata features to model item similarity. In the context of board games, this approach is especially relevant because games often contain rich descriptive information, including descriptions, mechanics, categories, themes, and complexity indicators. However, content-based systems may be limited when item metadata does not fully capture user experience or subjective preference.

Hybrid recommendation systems combine multiple recommendation strategies to improve robustness and recommendation quality. Burke [4] presents hybrid recommender systems as a way to reduce the weaknesses of individual recommendation approaches by combining different sources of evidence. More recent surveys also show that hybrid approaches remain important across different application domains because they can integrate collaborative, content-based, and contextual signals [5]. This motivates the design of the proposed system, which combines content similarity, sentiment information, and popularity signals.

Sentiment-aware recommendation systems incorporate user opinions from reviews to improve recommendation relevance. Wang et al. [6] demonstrate the usefulness of sentiment information in recommender systems by integrating sentiment with contextual recommendation signals. Da’u and Salim [7] also show that sentiment-aware models can enhance recommendation performance by capturing opinion-oriented information from review text. For board game recommendations, sentiment is valuable because user reviews often describe enjoyment, difficulty, replayability, and perceived game quality beyond numerical ratings.

Explainability has also become an important direction in recommender system research. Zhang and Chen [8] argue that explainable recommendation systems can improve user trust, transparency, and decision support. Chatti et al. [9] further discuss the role of visualization in recommendation explainability. This is important for board game discovery because users may want to understand why a specific game is recommended, such as shared mechanics, similar themes, positive sentiment, or popularity.

Evaluation is another critical aspect of recommender system development. Ranking-based metrics such as Recall@K and NDCG@K are commonly used to assess whether relevant items appear near the top of a recommendation list. Radlinski and Craswell [10] compare the sensitivity of information retrieval metrics, while Sakai and Kando [11] discuss evaluation metrics under incomplete relevance assessments. These studies support the use of ranking-oriented evaluation for recommendation tasks where the order of suggested items is important.

Based on the reviewed literature, several gaps remain relevant to board game recommendation. Collaborative filtering methods may struggle under sparse user interaction data, while pure content-based systems may ignore subjective user sentiment. Popularity-based recommendations are simple and robust but may overemphasize mainstream games. Existing studies also often focus on algorithmic accuracy without providing flexible query modes and interpretable explanations for end users. Therefore, this paper proposes an explainable hybrid board game recommendation system that integrates content similarity, sentiment signals, and popularity information while supporting title-based, trait-based, and combined recommendation modes.

III. METHODOLOGY

This section describes the dataset, feature engineering process, recommendation architecture, and evaluation pipeline used in the proposed hybrid board game recommendation system.

A. Dataset Description

This study utilized publicly available board game data collected from the BoardGameGeek (BGG) platform. The dataset contains board game metadata, user ratings, and textual user reviews. After preprocessing and filtering, the final dataset included over 21,000 board games and millions of user rating interactions.

The dataset preprocessing stage involved removing incomplete records, filtering games with insufficient metadata, and cleaning textual review content. Board game attributes such as categories, mechanics, themes, descriptions, complexity ratings, and user rating statistics were retained for recommendation modeling.

User rating data was restricted to games in the catalog, leaving 18,744,924 ratings (approximately 18.7 million) from 410,471 users.

Proposed Hybrid Board Game Recommendation Framework

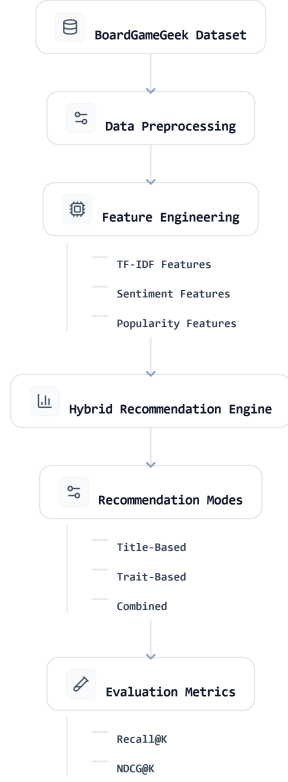


Fig. 1. Proposed Hybrid Board Game Recommendation Framework

The evaluation subset was constructed with a holdout strategy applied to this interaction data. Users with fewer than six rated games were excluded, and each remaining user’s ratings were split into seed-game inputs and held-out relevance targets, as detailed in Section III-G.

Textual review data was used for sentiment analysis, while game descriptions, categories, and mechanics were used for content-based similarity modeling. The processed dataset was then used to support title-based, trait-based, and combined recommendation experiments.

B. Feature Engineering

Feature engineering was performed to extract meaningful recommendation signals from board game metadata and review information. The proposed system combines content similarity features, sentiment features, and popularity indicators.

1) *Content Representation*: Content-based similarity was computed from three complementary representations of each game:

- **Description TF-IDF**: unigram and bigram TF-IDF vectors (20,000 features) of the BGG game description, capturing keyword-level similarity.
- **Attribute vectors**: L2-normalized binary indicators of the game’s BGG mechanics, themes, and subcategories, capturing structural gameplay similarity.
- **Semantic embeddings**: the game name and description encoded with the all-MiniLM-L6-v2 sentence-transformer (first 256 tokens), capturing higher-level semantic relationships that keyword overlap misses.

Cosine similarity was computed within each representation, and each game’s 50 nearest neighbours were stored as a sparse matrix.

2) *Trait Extraction*: Trait-based recommendation utilized four structured BGG attributes: categories, mechanics, families, and publishers. For each attribute, the overlap score of a candidate game is the fraction of the user’s requested values that the game carries. A small hand-written expansion table maps broad keywords to related BGG terms (for example, “strategy” expands to categories such as Economic and Territory Building and mechanics such as Hand Management and Tile Placement), so users can describe preferences in everyday words.

Trait matching was implemented as a soft overlap mechanism rather than a strict binary filter: games sharing a larger proportion of the requested traits receive higher trait scores. Complexity (BGG average weight) is not part of the trait score; it is exposed as an optional post-ranking filter with three bands (low ≤ 2.0 , medium 2.0–3.0, high > 3.0).

3) *Sentiment Analysis*: User review sentiment was incorporated as an additional recommendation signal. Sentiment scores were generated using the HuggingFace transformer pipeline with the DistilBERT SST-2 sentiment classification model applied to textual user reviews collected from BoardGameGeek.

Scoring every comment in the review corpus (about 26 million ratings) on a CPU was impractical, so up to ten non-empty comments per game were sampled uniformly at random, giving 209,162 comments across 21,581 catalog games; comments were truncated to 128 tokens. The classifier returns a label and the confidence in that label, so each prediction was converted to the probability that the review is positive (p for a POSITIVE label, $1-p$ for a NEGATIVE one) before averaging per game. Games with fewer than three scored comments were given no score and fall back to the catalog mean, which leaves 21,510 games with a game-specific sentiment value. The per-game values were min-max normalized before integration into the hybrid score.

4) *Popularity Signal*: Popularity-based information was incorporated using aggregate BGG rating statistics. In the title-based score, popularity P is the game’s number of user ratings, min-max normalized over the catalog. The trait-based and combined modes additionally use the Bayesian average

rating (R) and the log-scaled rating count (V), both min-max normalized over the candidate set, as small tie-breaking terms.

Popularity features were normalized prior to score combination so that they stay on the same $[0, 1]$ scale as the similarity signals.

C. Score Normalization

The recommendation framework combines multiple heterogeneous signals, including content similarity, trait overlap, sentiment scores, and popularity statistics. Because these components operate on different numerical scales, normalization was applied before score aggregation to ensure balanced contribution within the hybrid recommendation framework.

For a query with one or more seed games, a candidate's similarity in each matrix is the mean of its stored similarities to the seeds that list it among their 50 neighbours, and zero if none do. The keyword signal T is the average of the description and attribute similarities, and the semantic signal E is the embedding similarity; each is then min-max normalized over the whole catalog. Trait overlap scores already lie in $[0, 1]$ because they are proportions of matched attributes.

Popularity (rating counts) and aggregated sentiment scores were also min-max normalized prior to integration into the final recommendation formulas.

D. Hybrid Recommendation Formula

Each recommendation mode combines the normalized signals with its own fixed weights. The title-based score for a candidate game is

$$S_{title} = 0.35 T + 0.35 E + 0.10 S + 0.20 P \quad (1)$$

where T is the keyword (description and attribute) similarity, E the semantic embedding similarity, S the normalized sentiment score, and P the normalized rating count.

The trait overlap score combines the four attribute overlaps (categories O_{cat} , mechanics O_{mech} , families O_{fam} , publishers O_{pub}):

$$S_{trait} = 0.40 O_{cat} + 0.35 O_{mech} + 0.15 O_{fam} + 0.10 O_{pub} \quad (2)$$

and the trait-based mode ranks candidates by $0.70 S_{trait} + 0.15 R + 0.10 V + 0.05 S$. The combined mode merges both candidate sets and ranks by

$$S_{comb} = 0.65 S_{title} + 0.35 S_{trait} + 0.08 R + 0.04 V + 0.03 S \quad (3)$$

where a candidate missing from one side contributes zero for that side.

The weights were set manually rather than learned or tuned on the evaluation data. Similarity terms receive most of the weight because finding games similar to the user's stated taste is the primary objective of this discovery-oriented framework; popularity and sentiment act as secondary signals.

E. Recommendation Modes

The proposed framework supports three recommendation modes designed for different board game discovery scenarios.

1) *Title-Based Recommendation*: The title-based mode generates recommendations using similarity relationships derived from a selected seed board game. This mode is driven by content similarity, supported by sentiment and popularity.

2) *Trait-Based Recommendation*: The trait-based mode allows users to explore games using gameplay-related characteristics (categories, mechanics, families, and publishers), optionally restricted to a complexity band. This mode was designed to support exploratory discovery scenarios where users may not have a specific seed game in mind.

3) *Combined Recommendation*: The combined recommendation mode integrates both seed-game similarity and trait-based filtering to provide guided discovery recommendations. This mode was intended to balance semantic similarity with user-selected gameplay preferences, enabling more flexible exploration of the board game catalog.

F. Post-Ranking Rules

Two rules are applied after ranking and are therefore not part of any score. First, the optional difficulty filter keeps only games in the selected complexity band. Second, a same-series cap keeps at most two results that share a BGG series tag (family tags beginning with "Game:" or "Series:") with a seed game. Without it, pure similarity fills the list with a seed's own editions and expansions (for example, six Ticket to Ride variants for a Ticket to Ride seed), which defeats discovery. A capped series is named in the explanation, so the rule remains visible to the user.

G. Evaluation Strategy

The recommendation framework was evaluated using a holdout-based recommendation protocol constructed from the BoardGameGeek user rating data (18.7 million ratings on catalog games).

Of the 259,737 users with at least six rated games, the first 300 in the order produced by grouping the ratings by username were evaluated. For each user, the first three rated games in the order they appear in the ratings file were used as seed games, and the next three were withheld as ground-truth relevance targets. Because the ratings file is grouped by game with the most-rated games first, both seeds and held-out games are skewed towards popular titles. This should be kept in mind when comparing against a popularity baseline.

Recommendation quality was evaluated using Recall@K and NDCG@K metrics at multiple cutoff levels. A recommendation was considered relevant if it matched one of the user's held-out games; any rated game counts as relevant, regardless of the rating value. For the trait-based and combined modes, the trait query was derived automatically from the seeds (their two most frequent categories and two most frequent mechanics). All modes were evaluated through the same engine code that powers the deployed application.

Recall@K was used to measure the proportion of relevant games successfully retrieved within the top-K recommendations, while NDCG@K was used to evaluate ranking quality by assigning higher importance to correctly ranked recommendations appearing earlier in the recommendation list.

H. System Implementation

The recommendation system was implemented using Python-based data science tools and libraries. Data preprocessing and analysis were performed using Pandas and NumPy, while TF-IDF vectorization and similarity calculations were implemented using Scikit-learn. The user interface prototype was developed using Streamlit to provide interactive recommendation exploration and explainable recommendation outputs.

The system architecture was designed to support modular recommendation experiments and evaluation workflows.

IV. EVALUATION

This section presents the experimental setup, evaluation metrics, baseline methods, and performance analysis used to evaluate the proposed hybrid board game recommendation system.

A. Explainability Example

To improve recommendation transparency, the proposed system attaches an explanation to every recommendation. The explanation lists the concrete BGG attributes behind the match: the categories, mechanics, families, and publishers that the recommended game shares with the seed games (title-based mode) or with the requested traits (trait-based and combined modes). Alongside it, the interface shows the component scores (S_{title} , S_{trait} , and the final score), the Bayesian average rating, and the rating count, so users can see how much of a ranking comes from similarity and how much from the supporting signals. The interface also shows which catalog game each typed title was matched to, because title lookup falls back to substring matching.

Table I reproduces, verbatim, the explanations the system produced for three example queries.

TABLE I
VERBATIM EXPLANATIONS PRODUCED BY THE SYSTEM (PUBLISHER
LISTS ABBREVIATED)

Query → Game	System explanation
Splendor + Azul → Sultaniya	Categories: Puzzle Mechanics: Pattern Building, Set Collection, Tile Placement
Splendor + Azul → Azul: Summer Pavilion	Categories: Abstract Strategy, Puzzle Mechan- ics: Card Drafting, End Game Bonuses, Pattern Building, Set Collection, Tile Placement, Turn Order: Claim Action Families: Game: Azul Publishers: Broadway Toys LTD, ...
Ticket to Ride → Railroad Ink: Deep Blue Edi- tion	Categories: Trains Mechanics: Network and Route Building Publishers: Edge Entertainment, Galápagos Jogos
Horror + Mystery (traits) → Mys- terium	Categories: Deduction, Horror Mechanics: Co- operative Game, Deduction

B. Evaluation Metrics

To evaluate recommendation quality, this study utilizes ranking-based evaluation metrics commonly used in recommender systems research, namely Recall@K and Normalized Discounted Cumulative Gain (NDCG@K).

Recall@K measures the proportion of relevant items successfully retrieved within the top-K recommendations. This metric is useful for evaluating whether the recommendation system is capable of retrieving relevant board games for users.

The Recall@K metric is defined as follows:

$$Recall@K = \frac{\text{Number of Relevant Recommended Items}}{\text{Total Number of Relevant Items}} \quad (4)$$

NDCG@K evaluates ranking quality by assigning higher importance to relevant items appearing near the top of the recommendation list. Unlike Recall@K, NDCG@K considers recommendation ordering and ranking position.

The Discounted Cumulative Gain (DCG) is computed as:

$$DCG@K = \sum_{i=1}^K \frac{rel_i}{\log_2(i+1)} \quad (5)$$

The normalized version is computed as:

$$NDCG@K = \frac{DCG@K}{IDCG@K} \quad (6)$$

where $IDCG@K$ represents the ideal discounted cumulative gain.

These metrics were selected because recommendation ranking quality is important in board game discovery scenarios, where users typically interact with only the top portion of recommendation results.

C. Experimental Setup

The recommendation experiments were conducted using the processed BoardGameGeek dataset. The dataset was divided into recommendation input data and evaluation targets for ranking analysis.

The following models were evaluated on the same 300 users:

- Title-based recommendation,
- Trait-based recommendation,
- Combined hybrid recommendation,
- a popularity baseline,
- a content-only baseline, and
- a title-based variant without sentiment (ablation).

Recommendation results for the three proposed modes were generated using the scoring framework described in Section III, with no difficulty filter applied. Each evaluation is a single run without repeated sampling or significance testing.

1) *Popularity-Based Baseline*: The popularity-based baseline recommends the games with the most user ratings, excluding the user's seed games, independent of the seeds' content. This baseline represents a simple recommendation strategy commonly used in many recommendation platforms.

2) *Content-Based Baseline*: The content-based baseline is the title-based scorer with the sentiment and popularity weights set to zero ($0.5T + 0.5E$). It measures what content similarity alone achieves and therefore what the hybrid components add.

3) *Sentiment Ablation*: To isolate the sentiment signal, the title-based scorer was also evaluated with only the sentiment weight set to zero ($0.35T + 0.35E + 0.20P$).

D. Experimental Results

Table II reports the main protocol and Fig. 2 shows Recall@10 and NDCG@10.

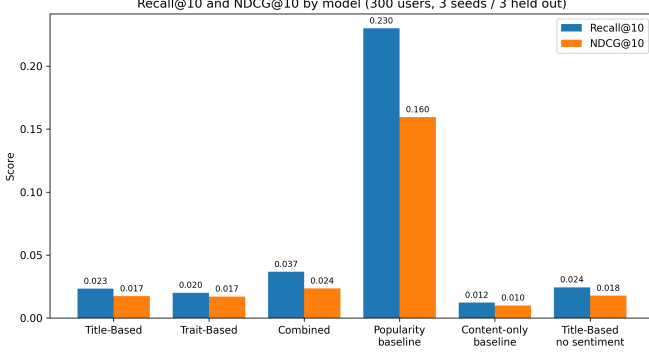


Fig. 2. Recall@10 and NDCG@10 of the proposed modes, the baselines, and the sentiment ablation (300 users)

TABLE II
RECOMMENDATION PERFORMANCE (300 USERS, 3 SEEDS / 3 HELD OUT)

Model	Recall@5	NDCG@5	Recall@10	NDCG@10	Recall@20	NDCG@20
Title-Based	0.0133	0.0128	0.0233	0.0174	0.0411	0.0236
Trait-Based	0.0133	0.0140	0.0200	0.0171	0.0444	0.0256
Combined	0.0222	0.0168	0.0367	0.0235	0.0556	0.0302
Popularity baseline	0.1400	0.1190	0.2300	0.1595	0.3622	0.2068
Content-only baseline	0.0111	0.0095	0.0122	0.0100	0.0167	0.0116
Title-Based, no sentiment	0.0133	0.0128	0.0244	0.0178	0.0367	0.0222

The popularity baseline clearly outperformed all three proposed modes, reaching a Recall@10 of 0.230 against 0.037 for the best proposed mode (combined). This is a property of the protocol as much as of the models: 63.0% of all held-out games are among the 100 most-rated titles in the catalog, so a list of the most-rated games retrieves many of them by construction. Among the proposed modes, the combined mode performed best on every metric, and the title-based and trait-based modes performed similarly to each other.

TABLE III
LONG-TAIL ANALYSIS: HELD-OUT GAMES OUTSIDE THE 100 MOST-RATED (155 USERS)

Model	Recall@5	NDCG@5	Recall@10	NDCG@10	Recall@20	NDCG@20
Title-Based	0.0129	0.0082	0.0151	0.0092	0.0258	0.0124
Trait-Based	0.0054	0.0051	0.0183	0.0093	0.0344	0.0141
Combined	0.0194	0.0101	0.0215	0.0110	0.0398	0.0166
Popularity baseline	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Content-only baseline	0.0129	0.0090	0.0151	0.0100	0.0215	0.0122
Title-Based, no sentiment	0.0129	0.0082	0.0151	0.0091	0.0280	0.0129

Table III re-scores the same ranked lists against only the held-out games outside the 100 most-rated titles, for the 155 users who have at least one. The cut-off was fixed before the results were examined. Here the popularity baseline retrieves nothing, because its lists never leave the head of the

distribution, whereas the proposed modes do retrieve some less popular games that users went on to rate. The absolute values remain low, so this shows a qualitative difference in behaviour rather than strong long-tail accuracy.

The two comparison models isolate the supporting signals. Removing only the sentiment term changed the title-based Recall@10 from 0.0233 to 0.0244, a difference of a single hit across the 900 held-out games, so the sentiment signal has no measurable effect in this evaluation. Removing both sentiment and popularity (the content-only baseline) lowered Recall@10 to 0.0122, which is consistent with the popularity bias of the protocol: the popularity term, not content similarity, accounts for most of the title-based mode’s offline accuracy.

V. DISCUSSION

The central finding is that, on this dataset and protocol, popularity is a far stronger predictor of which games a user rates next than content similarity. This is consistent with the well-known popularity bias of offline recommender evaluation: BGG users rate many of the same widely played titles, and a protocol that holds out a user’s next ratings in a popularity-ordered file inherits that bias. An earlier version of this system appeared to reach a title-based Recall@10 of 0.20, but that figure was an artifact: its semantic-embedding and sentiment signals had been computed for only 20 very popular games, so the title-based mode effectively recommended those blockbusters to everyone. Once both signals were computed for the full catalog, the title-based mode recommended genuinely similar games and its accuracy fell to the level reported here.

The proposed modes are designed for discovery rather than for predicting the next popular game a user rates. The long-tail analysis supports this distinction, since only the similarity-based modes reach games outside the 100 most popular, but it does not show that they do so accurately. Combining seed similarity with explicit traits gave the best results among the proposed modes in the main analysis, which suggests that the user’s stated preferences add useful information to seed similarity.

The explanations remain the system’s main strength. Every recommendation is accompanied by the concrete categories, mechanics, families, and publishers it shares with the user’s input (Table I), the component scores, and the catalog game each typed title was matched to. Their main weakness is that long publisher lists, which mostly reflect localized editions, add noise to otherwise informative explanations.

Several limitations remain. First, the evaluation uses a single protocol with a popularity-ordered split, treats every rated game as relevant regardless of rating, uses only 300 users, and reports single runs without significance testing. Second, review sentiment is estimated from at most ten sampled comments per game, so individual values are noisy. Third, the weights were set manually and were not tuned. Fourth, the semantic embeddings include the game name, so games with similar names receive a small similarity boost. Finally, the system does not use collaborative filtering or personalized user modelling.

Future work should evaluate the system with a randomized split and a rating threshold for relevance, report confidence intervals, and, most importantly, run a user study that measures discovery and the perceived usefulness of explanations directly. Learning the weights on a validation split, re-ranking for diversity beyond the series cap, and combining the explainable content signals with a collaborative component are natural extensions.

VI. CONCLUSION

This study proposed an explainable, multi-mode board game recommendation framework that combines content similarity, gameplay trait matching, review sentiment, and popularity signals, and that explains every recommendation through the concrete attributes behind it. Evaluated through the same code that powers the deployed application, the three proposed modes were clearly outperformed by a popularity baseline on a holdout protocol dominated by popular games (Recall@10 of 0.037 for the best mode against 0.230), while reaching a small share of less popular held-out games that the popularity baseline never recommends.

The results do not support claims of superior retrieval accuracy. They support a narrower conclusion: a lightweight, fully interpretable content-based framework can surface relevant games beyond the most popular titles and explain why, but its usefulness for discovery must be demonstrated with user studies and less popularity-biased evaluation rather than with next-rating prediction alone.

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